

Sistem Terdistribusi

IF2228

01: Pengantar Perkuliahan

Agustus 2026

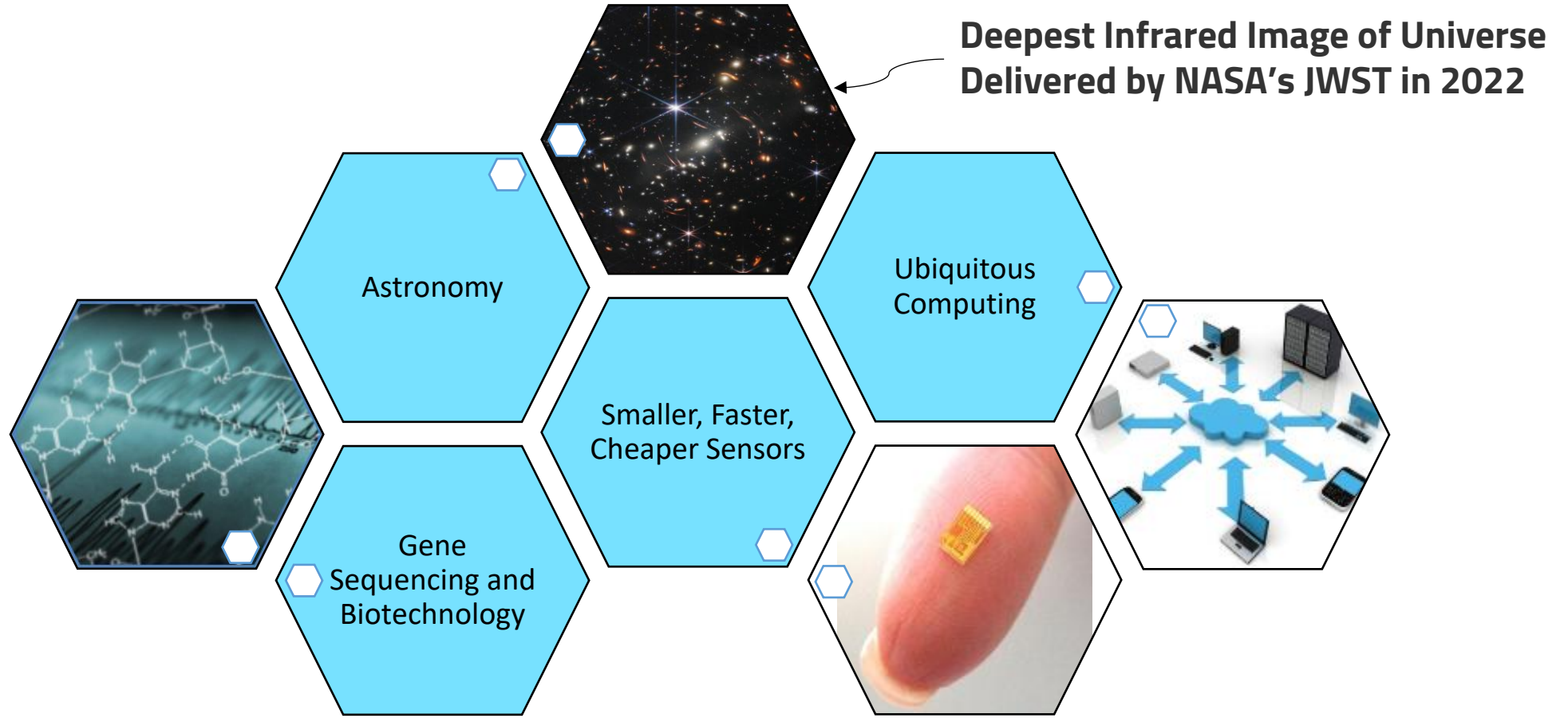
Sistem Terdistribusi

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Pengampu

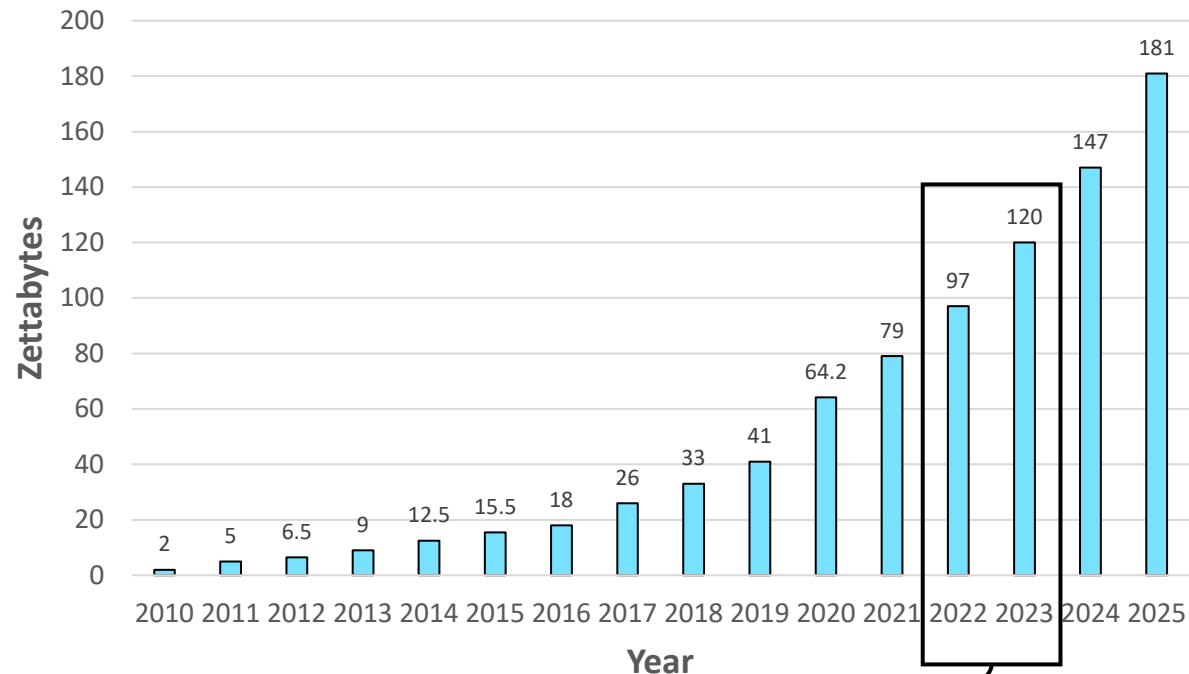
Yoga Dwitya P.

On the Verge of A Disruptive Century : Breakthroughs



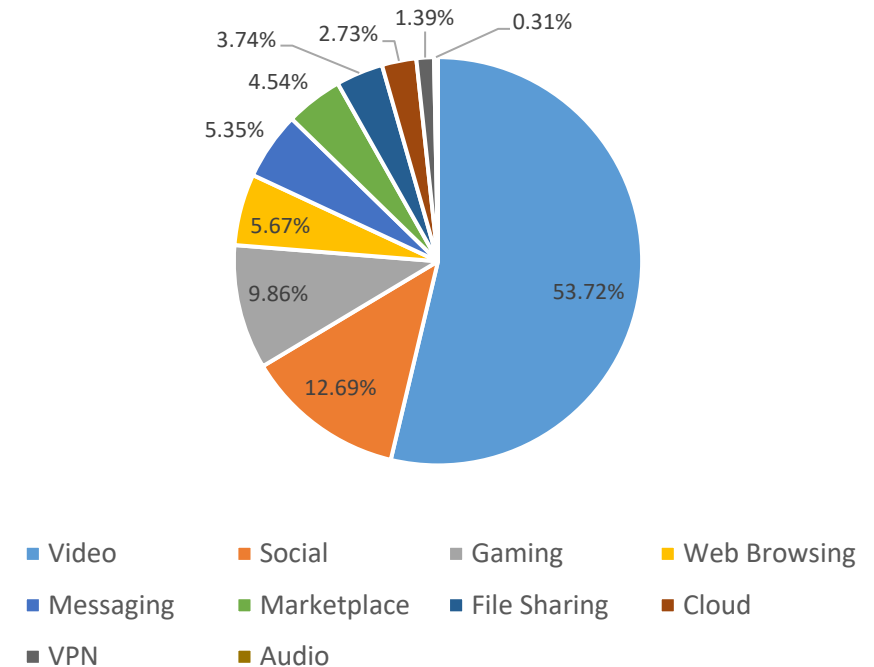
A Common Theme is Data

Data Generated Per Year



~70% of the world's data was generated only over the past two years

Proportion of Internet Data Traffic



We Live in a World of Data...

- A few examples (*all per day*):
 - 333.22 billion emails sent
 - 129.89 billion crypto purchased
 - 8.5 billion Google searches
 - 1.44 billion hours streamed
 - 637.92 million USD spent on Amazon
 - 150.62 million hours spent in Zoom meetings
 - 110.02 million USD spent on DoorDash
 - ...

What Do We Do With Data?



Store



Share



Access



Process



Encrypt



.... and
more!

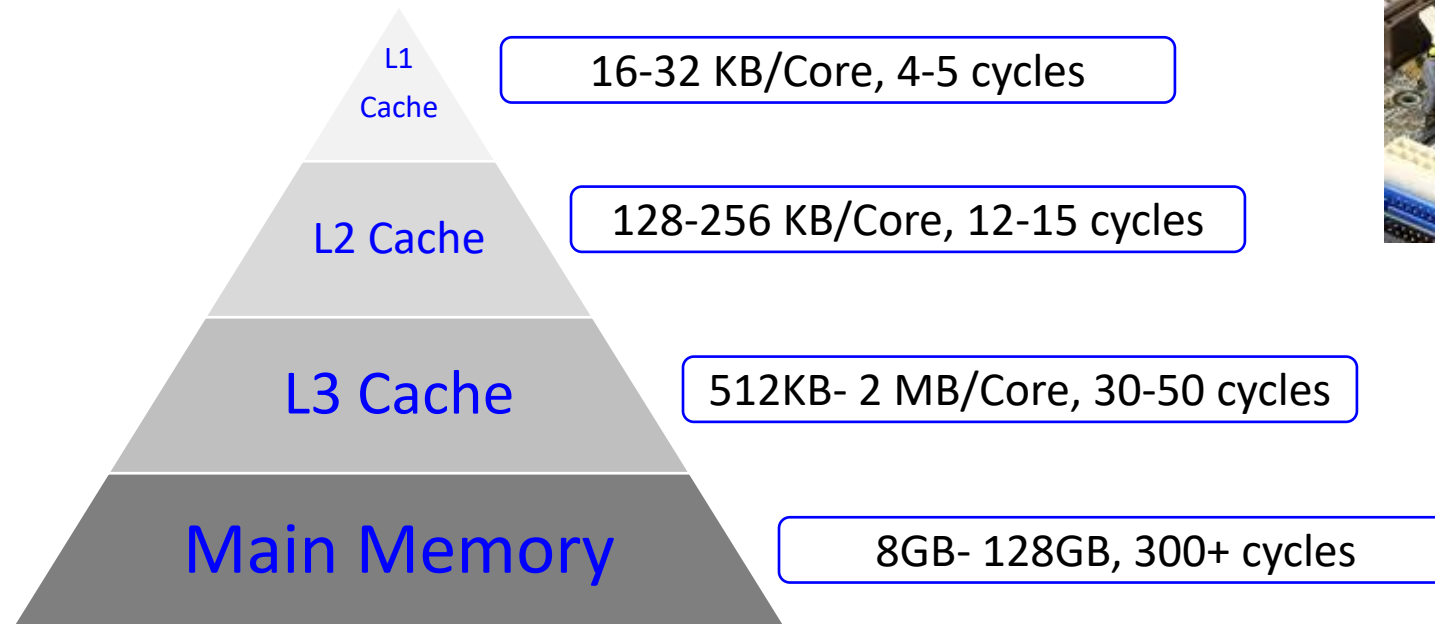
How to Store and Process Data at Scale?

- A system can be scaled:
 - Either *vertically* (or *up*)
 - Can be achieved by hardware upgrades (e.g., faster CPU, more memory, and/or larger disk)
 - And/Or *horizontally* (or *out*)
 - Can be achieved by adding more machines

Vertical Scaling

- **Caveat:** Individual computers can still suffer from *limited resources* with respect to the scale of today's problems

1. Caches and Memory:

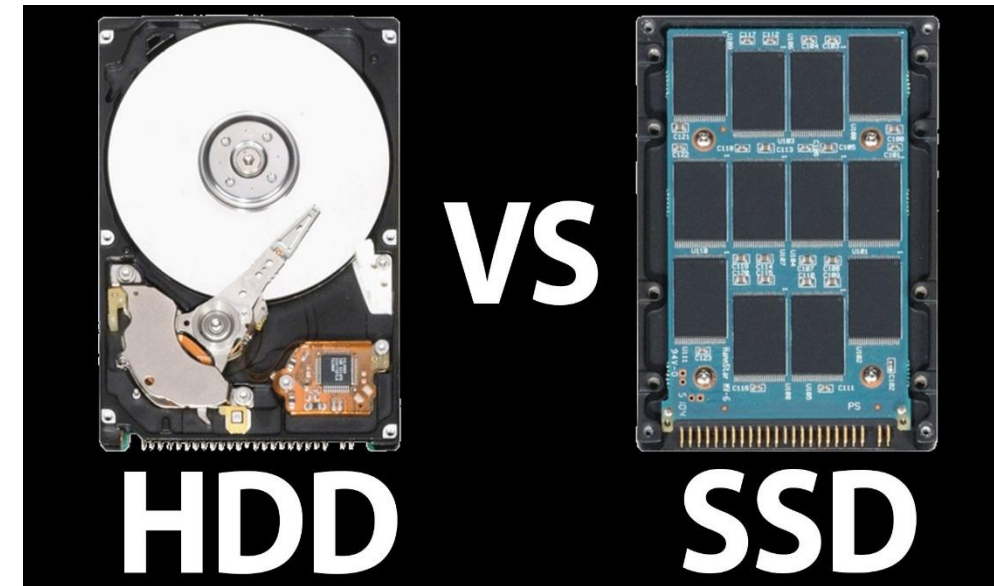


Vertical Scaling

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2. Disks-- some advancements, but still:

- Limited capacity
- Limited number of channels
- Limited bandwidth

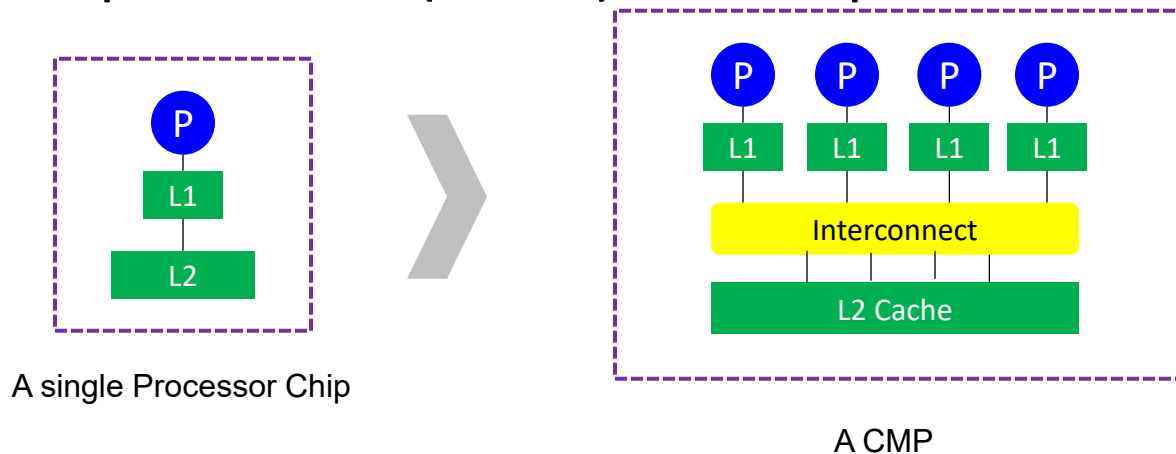


Vertical Scaling

- **Caveat:** Individual computers can still suffer from *limited resources* with respect to the scale of today's problems

2. Processors:

- Moore's law still holds
- Chip Multiprocessors (CMPs) are ubiquitous

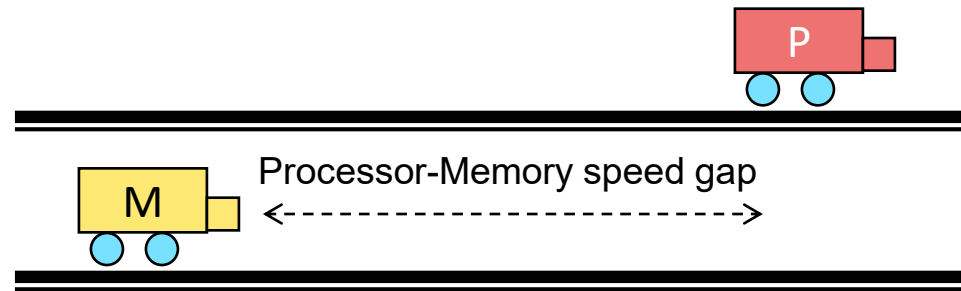


Vertical Scaling

- **Caveat:** Individual computers can still suffer from *limited resources* with respect to the scale of today's problems

2. Processors:

- But up until a few years ago, CPU speed grew at the rate of 55% annually, while the memory speed grew at the rate of only 7%



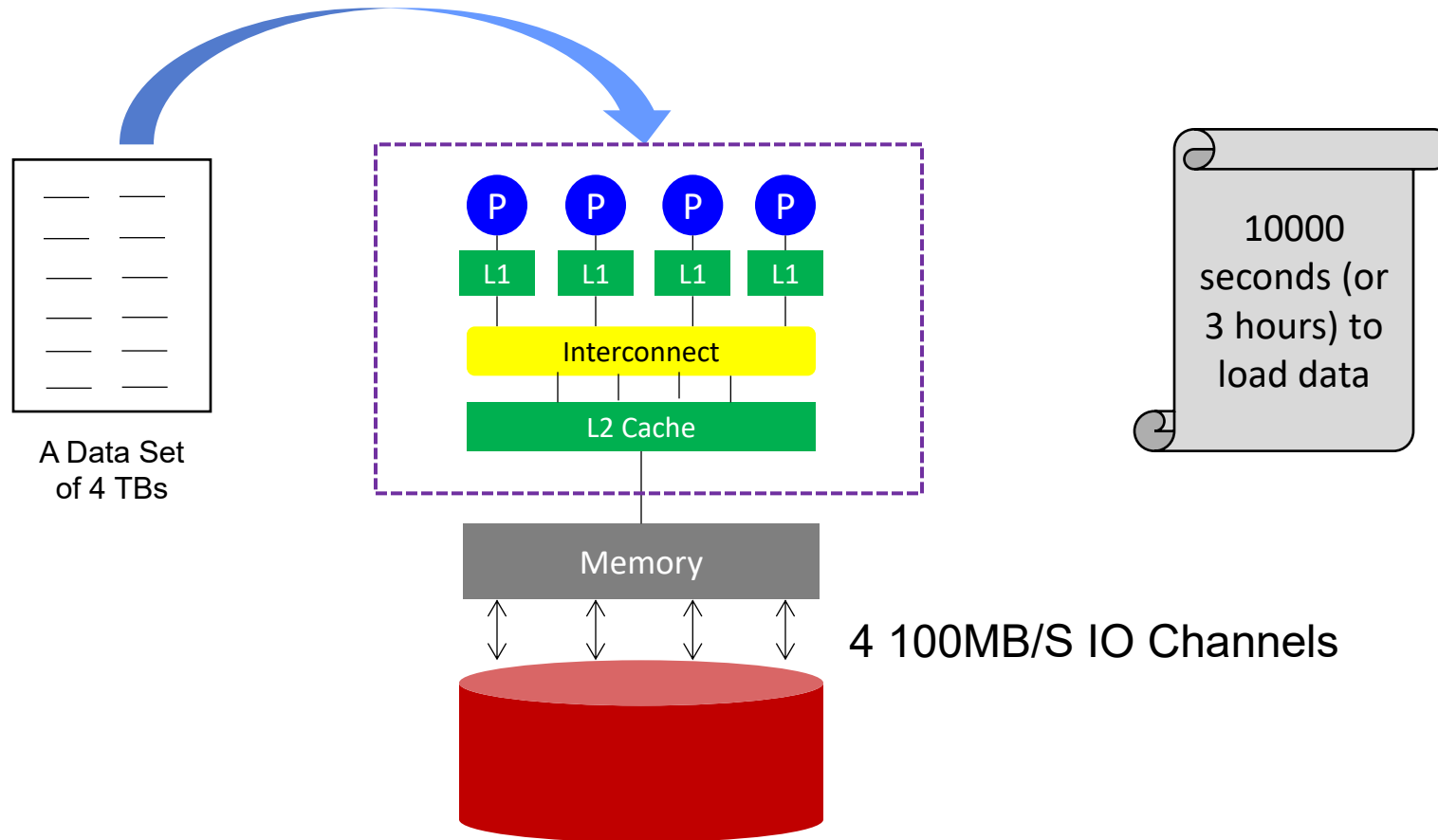
Vertical Scaling

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2. Processors:

- But up until a few years ago, CPU speed grew at the rate of 55% annually, while the memory speed grew at the rate of only 7%
- Even if 100s or 1000s of cores are placed on a CMP, it is a challenge to deliver input data to these cores fast enough for processing

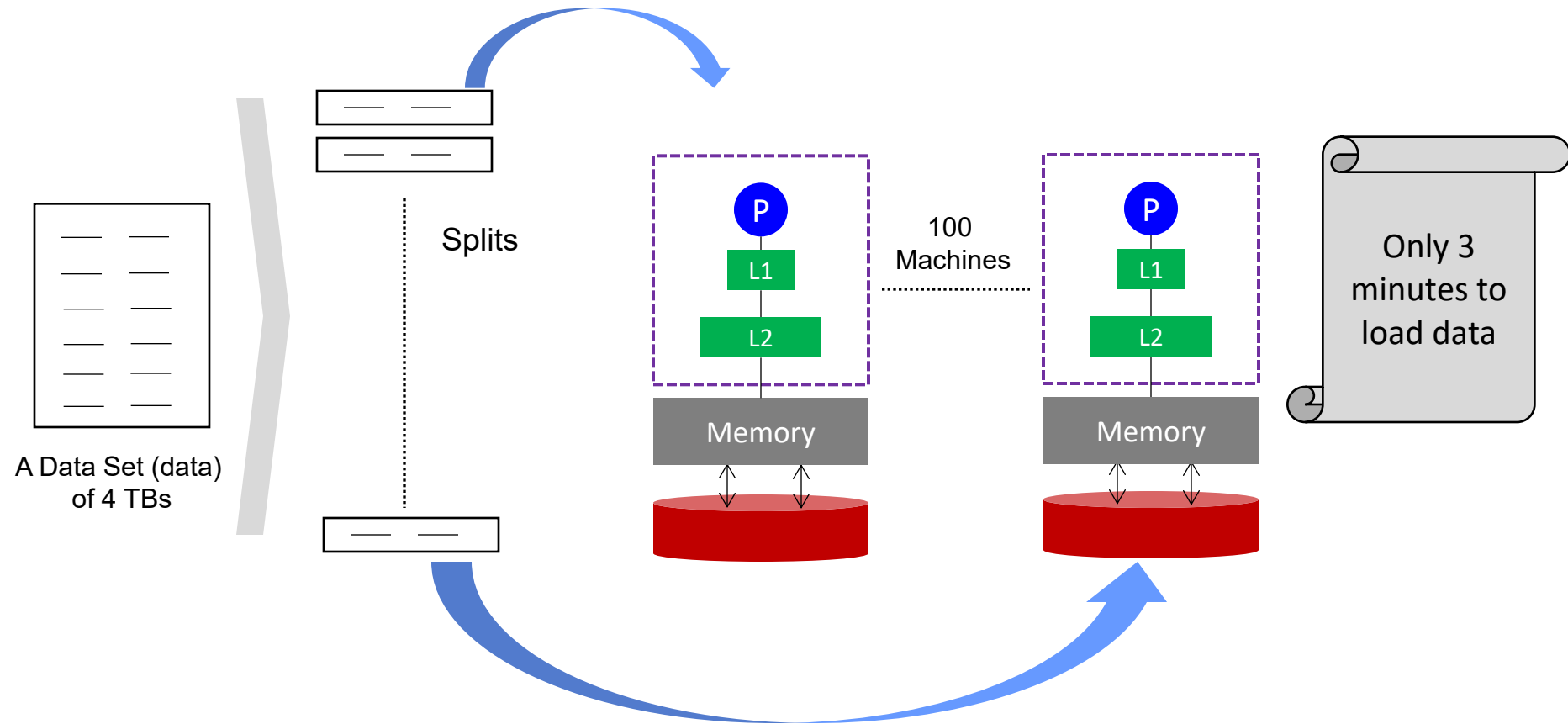
Vertical Scaling



How to Store and Process Data at Scale?

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Horizontal Scaling



Requirements

- But, this necessitates:
 - A way to express the problem in terms of parallel processes and execute them on different machines (*Programming and Concurrency Models*)
 - A way to organize processes (*Architectures*)
 - A way for distributed processes to exchange information (*Communication Paradigms*)
 - A way to locate and share resources (*Naming Protocols*)

Requirements

- But, this necessitates:
 - A way for distributed processes to cooperate, synchronize with one another, and agree on shared values (*Synchronization*)
 - A way to reduce latency, enhance reliability, and improve performance (*Caching, Replication, and Consistency*)
 - A way to enhance load scalability, reduce diversity across heterogeneous systems, and provide a high degree of portability and flexibility (*Virtualization*)
 - A way to recover from partial failures (*Fault Tolerance*)

So, What is a Distributed System?

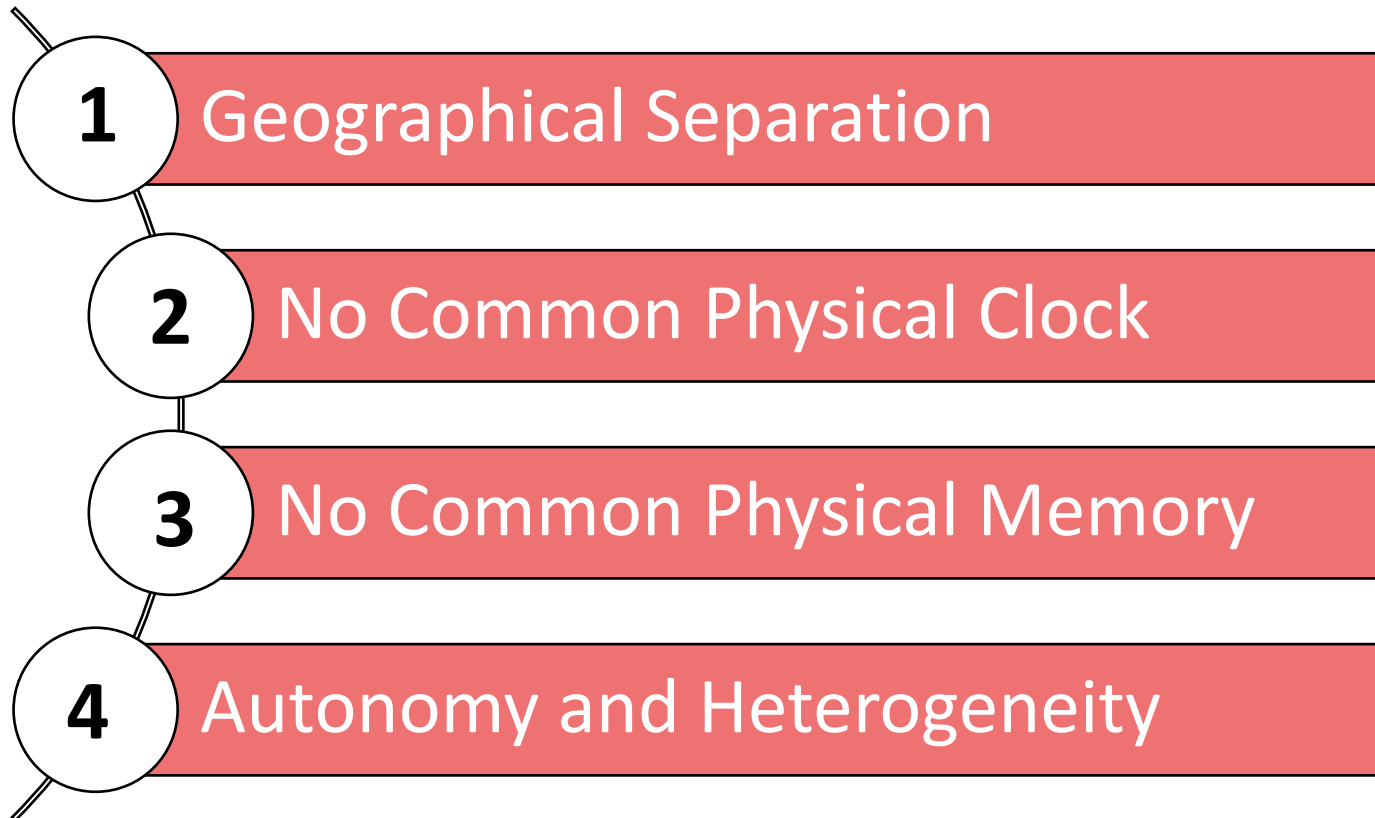
A distributed system is:

A collection of independent computers that appear to its users as a single coherent system

One in which components located at networked computers communicate and coordinate their actions only by passing messages

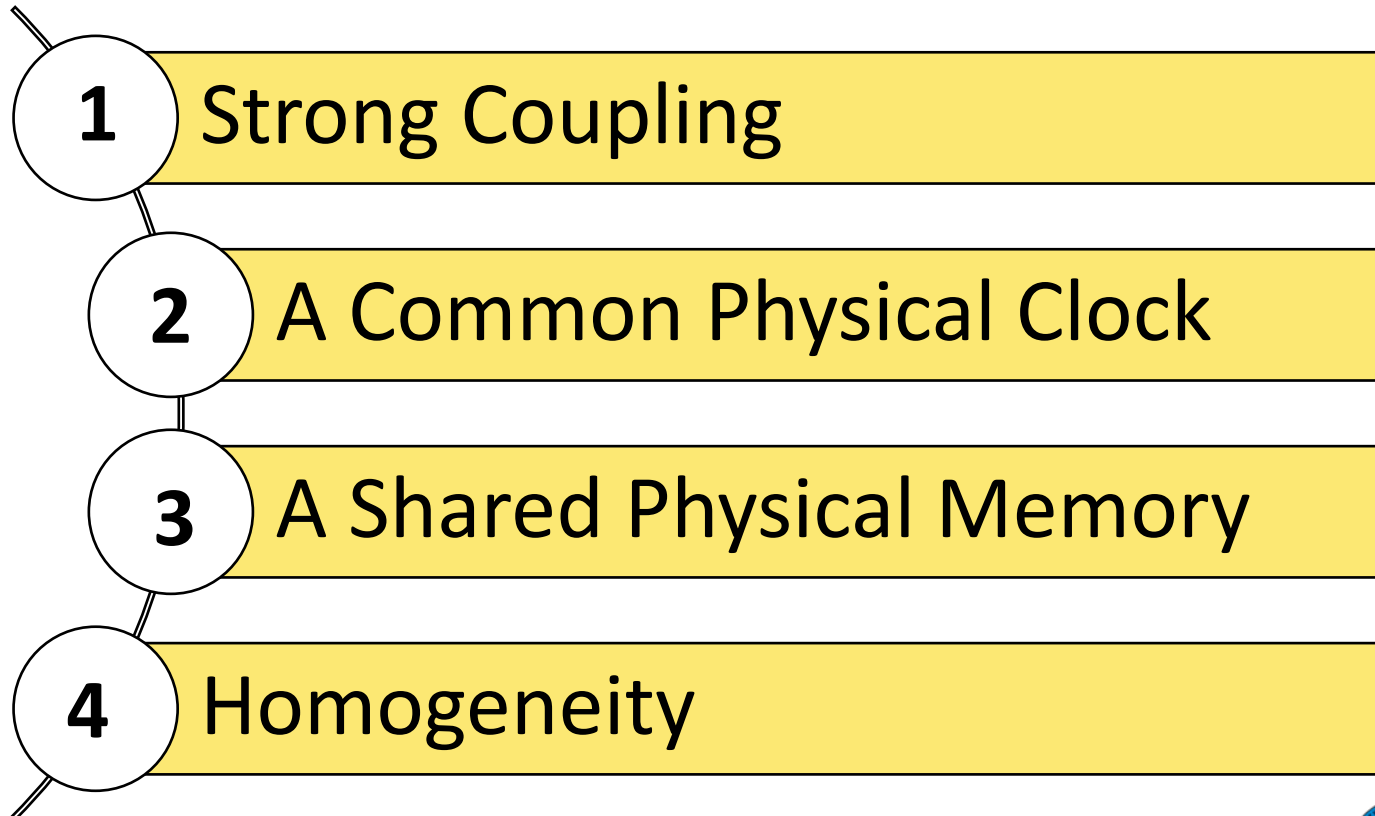
Features

- Distributed Systems imply *four* main features:



Parallel vs. Distributed Systems

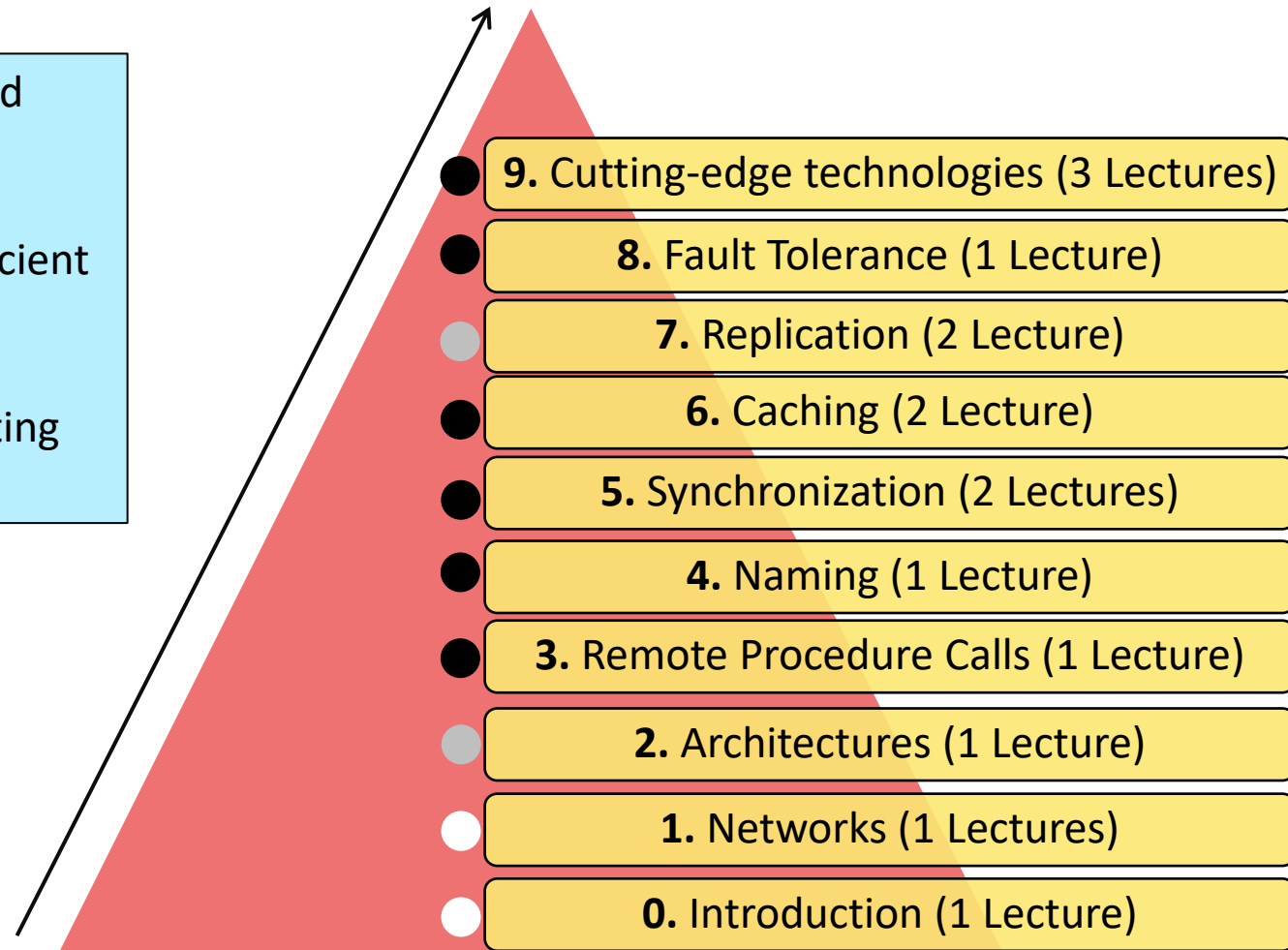
- Distributed systems contrast with parallel systems, which entail:



Kuliah ini...

Introduction to Distributed Systems

- **Considered:** a reasonably critical and comprehensive perspective.
- **Thoughtful:** Fluent, flexible and efficient perspective.
- **Masterful:** a powerful and illuminating perspective.



Course Objectives

The course aims at providing an in-depth understanding and hands-on experience on the

Architectures, communication paradigms, naming, and synchronization

Principles on which distributed systems are *based*

Principles on which distributed systems are *optimized*

Classical and modern distributed system programming models

Cutting-edge new technologies

Cloud Computing, Blockchain, and distributed LLMs

MPI, MapReduce, Spark, Snowflake, and Ray

Replication and fault tolerance



Assessment Methods

- How do we measure learning?

Type	Number	Weight
Project	4	40%
Exam	2	30% (Midterm: 10% & Final: 20%)
Problem Set	5 or 6	12%
Quiz	2	10%
Attendance & Participation	40	8%

- ***To pass the course, you need to pass its hands-on (i.e., projects) and the theory components***



Rules on the Projects

- For all the projects except the final one, the following rules apply:
 - If you submit one day late, 25% will be deducted from your project score
 - If you are two days late, 50% will be deducted
 - The project will not be graded (and you will receive a zero score) if you submit more than two days late
 - You have a ***3-grace-day quota***

Next Lecture...

- Networking

Questions?